4.10.1 Extinguishing hydrogen fire

4.10.1.1 As mentioned in paragraph 4.6, for flammable products the Code already requires elimination of sources of ignition, including use of electrical installations of appropriate types in order to minimize the risk of fire and explosion. No special requ irement is considered necessary with regard to ignitability of hydrogen.

4.10.1.2 Furthermore, with regard to the wide range of flammable limits of hydrogen, the increased quantities of carbon dioxide as a fire -extinguishing medium should be specified as mentioned in paragraph 4.7. No additional special requirement is considered to be necessary with regard to the wide range of flammable limits of hydrogen.

4.10.2 Disposal of cold hydrogen gas

The wide flammability range makes disposal of cold hydrogen gas a major hazard. Cold
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SOURCE: MSC.565(108).pdf
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https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.565(108).pdf
4.10.2 Disposal of cold hydrogen gas

The wide flammability range makes disposal of cold hydrogen gas a major hazard. Cold plumes of released hydrogen may impede adequate dilution of hydrogen down to below 4% and may lead to flash-back to the vent from distant ignition sources outside safety -controlled areas. The low ignition energy and wide flammable range may present significant challenges.

4.11 Prevention of dangerous purging operation

4.11.1 During cargo operations for maintenance, pipes and tanks should be purged with an inert gas or inert gases as illustrated in the figure below. For safety, due consideration should be given to temperature and boiling points of the inert gases. Residual pockets of hydrogen or the purge gas will remain in the enclosure if the purging rate, duration, or extent of mixing is too low. Therefore, reliable gas concentration measurements should be obtained at a number
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SOURCE: MSC.565(108).pdf
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too low. Therefore, reliable gas concentration measurements should be obtained at a number of different locations within the system for suitable purges. Temperature should also be measured at a number of locations. Oxidizing agents may exist in a hydrogen containing equipment, specifically: air, cold box atmospheres containing air diluted with nitrogen, or oxygen-enriched air that can be condensed on process pipe work within the cold box in special circumstances.

4.11.2 There are special measures that may need to be put in place in order to mitigate the hazards, e.g. air should be eliminated by nitrogen purge prior to introduction of hydrogen into cargo piping or processing equipment. Nitrogen should then be eliminated by hydrogen purge, where there is a possibility of its solidification in the subsequent process.

Figure 1
References in part A

1 ISO/TR 15916:2015, Basic consideration for the safety of hydrogen systems (ISO)

SOURCE: MSC.565(108).pdf
URL:
https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.565(108).pdf
Figure 1
References in part A

1 ISO/TR 15916:2015, Basic consideration for the safety of hydrogen systems (ISO)

2 American Institute of Aeronautics and Astronautics, "Safety Standard for Hydrogen and Hydrogen Systems (Guide to Safety of Hydrogen and Hydrogen Systems)", 2005 (AIAA)

SOURCE: MSC.565(108).pdf
URL:
https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.565(108).pdf

3 NFPA 12: Standard on Carbon Dioxide Extinguishing Systems 2020 Edition (NFPA)

4 ISO/IEC 80079 -20-1:2017 Explosive atmospheres Part 20 -1: Material characteristics for gas and vapour classification Test methods and data

5 UN Recommendations on the Transport of Dangerous Goods Model Regulations, twenty-second revised edition

6 NFPA 2: Hydrogen Technologies Code 2016 Edition (NFPA)

7 IEC/ISO 31010: 2019 Risk management Risk assessment techniques

8 Cryogenics Safety Manual Fourth Edition (1998)

9 SAE ARP 5580 -2001 "Recommended failure modes and effects analysis (FMEA) practices for non-automobile applications"

10 National Institute of Standards and Technology (NIST) RefProp database

Part B
Cargo containment systems of independent cargo tanks using vacuum insulation

5 Additional requirements

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Part B
Cargo containment systems of independent cargo tanks using vacuum insulation

5 Additional requirements

5.1 Additional special requirements for cargo containment systems of independent cargo tanks using vacuum insulation are prescribed in table 4 and these special requirements should apply in addition to the requirements in table 2.

Table 4: Special requirements for cargo containment systems of independent cargo tanks using vacuum insulation

No.	Special requirement	Related hazard
B-1	The insulation performance of vacuum insulation of cargo containment system should be evaluated to the satisfaction of the Administration based on experiments, as necessary.	General (see 4.1 and 6.1)
B-2	Notwithstanding special requirement A -22, liquid and gas hydrogen pipes may pass through spaces constituting a part of a cargo containment system using vacuum insulation where the degree of vacuum is monitored.	Susceptibility to

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	cargo containment system using vacuum insulation where the degree of vacuum is monitored.	Susceptibility to leakage (see 4.4)
B-3	When selecting the most onerous scenario stipulated in special requirement A-24, the evaluation should include fire or loss of vacuum from the overall insulation system and should also consider the resulting magnitude of the heat flux in case of a single failure on the cargo containment system in each case.	High pressure hazard (see 4.8 and 6.2)

6 Additional special requirements to mitigate hazards of liquefied hydrogen

6.1 Hazards of liquefied hydrogen to be considered

6.1.1 In addition to 4.1.1, due consideration should be given to the possibility of untimely deterioration of insulation properties at the envisaged carriage temperatures of liquid hydrogen.

The vacuum insulation evaluation should be specified for the normal range or upper limit of cold
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The vacuum insulation evaluation should be specified for the normal range or upper limit of cold
vacuum pressure (CVP), and loss of vacuum should be defined with respect to this value.
Accordingly, effect of vacuum pressure should be taken into account at the time of design and
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SOURCE: MSC.572(110).pdf
URL:
https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.572(110).pdf
RESOLUTION MSC.572(110) (adopted on 26 June 2025)
AMENDMENTS TO CHAPTERS II-2 AND V OF THE
INTERNATIONAL CONVENTION FOR THE SAFETY OF LIFE AT SEA, 1974
RESOLUTION MSC.572(110)
(adopted on 26 June 2025)
AMENDMENTS TO CHAPTERS II-2 AND V OF THE
INTERNATIONAL CONVENTION FOR THE SAFETY OF LIFE AT SEA, 1974
THE MARITIME SAFETY COMMITTEE,
RECALLING Article 28(b) of the Convention on the International Maritime Organization
concerning the functions of the Committee,
RECALLING ALSO article VIII(b) of the International Convention for the Safety of Life at
Sea, 1974 ("the Convention"), concerning the amendment procedure applicable to the annex
to the Convention, other than to the provisions of chapter I,
HAVING CONSIDERED, at its 110th session, amendments to the Convention proposed and
circulated in accordance with article VIII(b)(i) of the Convention,
1 ADOPTS, in accordance with article VIII(b)(iv) of the Convention, amendments to
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SOURCE: MSC.572(110).pdf
URL:
https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.572(110).pdf
1 ADOPTS, in accordance with article VIII(b)(iv) of the Convention, amendments to
the Convention, the text of which is set out in the annex to the present resolution;
2 DETERMINES, in accordance with article VIII(b)(vi)(2)(bb) of the Convention, that
the said amendments shall be deemed to have been accepted on 1 July 2027, unless, prior to
that date, more than one third of the Contracting Governments to the Convention or